

Krishna Verma

Senior Staff Software Engineer

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✉ krisrajaryan27.github.io/Portfolio ✉ stackoverflow.com/users/6619434/krishna-verma ✉ linkedin.com/in/krishna-verma ✉ learningnirvana.home.blog ✉ github.com/krisrajaryan27

Summary

Senior Staff/Engineering Manager with extensive experience in backend and full-stack development, large-scale distributed systems, and event-driven microservices across energy, e-commerce, fintech, and SaaS domains, including leadership role seeks senior engineering leadership or staff/principal engineer roles focused on complex distributed systems, cloud-native platforms, and high-impact architectural initiatives.

Work experience

Senior Staff Software Engineer

Enphase Energy, Bangalore, India

Dec 2024 - Present

- Led the migration of a large monolithic legacy Ruby on Rails (ROR) codebase to an event-driven microservices architecture leveraging Java21, Spring WebFlux, Reactive Programming, MongoDB, MySQL, Cassandra, AWS stack, and Confluent Kafka with Docker and Kubernetes, grounded in domain-driven design principles; consolidation of ~50 VMs and achievement of approximately 70% infrastructure cost reduction.
- End-to-end design and implementation of processing of telemetry data from microcontrollers/PCUs and Envoy devices including high-level design (HLD), low-level design (LLD), system and cloud architectures, API development, and domain modeling for critical components managing over 20K reports per second across ~10 report types.
- Delivered high-impact solutions such as Audit Log Microservices, UDP Ping Service stabilization (SLA improvement from ~2% to ~99%), and Performance, Inventory, Notification, Configuration Reports, resulting in a 75% reduction in latency and a 60% increase in TPS; design and deployment of performance monitoring systems for solar panels, IQCPs, Envoys, and gateways ensuring reliable, efficient telemetry pipelines at enterprise scale.
- Technical leadership of a 25 developers' team to address diverse hardware programming challenges while meeting cross-functional requirements related to event handling and report generation: ownership and ramp-up across ENXT platforms including Task MS, Performance Reports, Inventory Reports, Notification Reports, and Config Reports with full lifecycle responsibility.
- Execution of key platform and data initiatives: design and delivery of Audit Log MS v1 APIs with HLD/LLD, large-scale MongoDB POC with architectural recommendations, Enterprise Redis integration to improve reliability and performance, and controlled rollout of migrations (1%, 20%, 100%) to reduce risk and ensure service stability.
- Implementation of Cassandra Events integration and support for Kafka Confluent migrations; delivery and support of IQ Power Match and major ecosystem features and standards including ENS 4.0, Sunspec, ITK, ENHO, GW3, HEMS, and NEM 3.0, enabling robust, scalable operation in complex hardware-oriented and regulatory environments.

Lead Software Engineer

Avataar.ai, Bangalore, India

Dec 2021 – Dec 2024

- Directed the development and integration of Avataar.ai's e-commerce platform with platforms such as BigCommerce, Shopify, Wix, and WooCommerce, enabling sellers to transform 2D products into immersive 3D experiences.
- Engineered the Avataar Engine leveraging MERN stack technologies alongside AWS Lambda, API Gateway, S3, SQS, and Google Analytics to deliver a scalable and robust platform architecture.
- Designed and deployed multiple microservices including adapter service, apheleia service (User and Team Management), Catalog Service, Translation Service, Work Item Service, Order Service, Cart Service, and Payment Service to power the Avataar ecosystem efficiently. Configured real-time alerts and monitored metrics through database events using Java 8, Spring Boot, gRPC, Postgres, and visualization tools like Grafana for enhanced operational insights.
- Developed a dynamic user interface using React for seamless interaction between users and the e-commerce platform; created reusable components to ensure design consistency and accelerated development timelines.
- Oversaw a team of 20 developers throughout the product lifecycle while delivering clear documentation to support efficient collaboration and maintaining high standards in code quality for error-free deployment.

Lead Software Engineer

Talentica Software, Pune, India

Dec 2018 – Sep 2021

- Designed and implemented microservices architecture for backend document retrieval, integrating financial search engine features into Alpha-sense using Java 8, Spring Boot, and AWS services including S3 and DynamoDB. Containerized applications with Docker and orchestrated deployment through Kubernetes while monitoring with Prometheus and Grafana.
- Developed the segment module on the Alpha-sense entity page to enable seamless feed integration of SEC and Moodys documents through APIs for various organizations, utilizing React, Redux, Typescript, Node.js, and implementing state management to handle complex user interactions effectively.
- Built from scratch a single sign-on feature for an existing recruitment platform using Java, Spring Boot, Struts 2.x, SAML 2.0, AngularJS, and integrated it with leading identity providers such as Microsoft ADFS, Ping Federate, Okta, OAuth, and SSOCircle.

- Engineered multiple microservices within the AtheerAir low-code platform to streamline manufacturing processes by enabling workflows for air forms creation and user management. Leveraged technologies including Java 8, Spring Boot, MongoDB along with AWS services (S3, Route53) while ensuring automation via Jenkins and observability through Datadog.
- Designed a custom customer dashboard portal in AtheerAir to monitor product pipeline statuses efficiently by integrating real-time updates with role-based access controls through comprehensive backend data flow systems. Enhanced application usability across systems by developing improved logging features coupled with single sign-on functionality that reduced redundant authentication steps for users and increased operational efficiency significantly.
- Led development of the entire backend module for Symphony, an end-to-end Dashboard using technologies such as Node.js, React, Redux, alongside API development with Spring Boot, Data JPA, and MongoDB for LEAP.
- Conducted technical interviews to recruit and identify suitable developers aligning with project requirements and organizational standards.
- Ensured adherence to best practices in software design, implementation, and deployment by optimizing systems for performance and maintainability.

Software Engineer

Mahindra Comviva Technologies, Bangalore, India

June 2015 – Dec 2018

- Analyzed telecom client requirements and delivered tailored solutions by evaluating, scoping, and finalizing project specifications.
- Designed, tested, and deployed scalable, modular, and high-performance telecom software products to meet industry standards.
- Acted as the primary contact for Loyalty Management Solution (LMS) of Mobilytix Suite deployed at HCPT (Indonesia), IRIS system at Banglalink (Bangladesh), and LMS for Viettel (Vietnam).
- Improved back-end performance by implementing a plugin model, leading to a 60% increase in transactions per second (TPS) and successfully replaced an 8year-old product at the Jakarta client site.
- Successfully deployed the IRIS solution in production environments across multiple regions including Bangladesh and Vietnam within tight timelines while resolving critical issues promptly during implementation.
- Developed Customer Care GUI, various types of reports, interface, and widget modifications to meet evolving customer requirements in dynamic project environment.

Education

Motilal Nehru National Institute of Technology

Bachelor of Technology

Electronics and Communications Engineering

CGPA- 8.32/10

Technical Skills

Backend Technologies - Java 21, Scala, Node.js, Python

Frameworks - Spring, Spring Web Flux, Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Spring Cloud, Struts 2.x, Servlet, Hibernate, Servlet, Express, MERN/MEAN stack, Reactive Programming, Django, Fast Api, Jupyter

Webservices - REST APIs, SOAP, gRPC, GraphQL

Unit Testing - JUnit5, Mockito, Spock (Groovy) Framework, Jest, Mocha, Sinon

Front End Technologies - HTML, CSS, JavaScript

Libraries and Frameworks - React, Angular 9, Redux, Typescript, Next.js, Nest.js

RDBMS - MySQL, Oracle 12g, Postgres, Cockroach DB

NoSQL - MongoDB, DynamoDB, Cassandra

Caching - Redis, Hazel cast, Memcache

Indexing - Apache Solr, Elastic Search

Message Queues - AWS SQS, Kafka (AWS, Confluent), RabbitMQ

Build Tools - Maven, Gradle, NPM, Yarn

AWS Services - EC2, ECS, EKS, Lambda, S3, SQS, SNS, SES, Dynamo, IAM, Route 53, Kinesis, Athena, Glue, CloudWatch

Version Control Systems - Git, Bitbucket/GitHub, SVN

Operating Systems - macOS, Linux, Ubuntu, Windows

Container and Orchestration - Docker, Kubernetes

Deployment Automation - Ansible, Chef

CICD - Jenkins, GitHub Actions

Metrics and Alerts Monitoring - Prometheus, Grafana, Datadog, New Relic, Splunk

Cloud Platforms - AWS, GCP, Azure

Agentic AI Tools - Cursor, Windsurf, Intellij JetBrains AI Assistant, Claude Code, Amazon Q